

Arif Agustyan

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EXPERIENCE

- Jul 2024 - Now **LinkAja (PT Fintek Karya Nusantara)** West Jakarta
Senior AI & ML Scientist Associate – Full Time
- Developed a **Generative RAG Chatbot for Codebase Visualization and Repository Q&A**. Designed a generative Retrieval-Augmented Generation (RAG) chatbot to **provide insights into a Golang code repository**. This chatbot **integrates reranking, embeddings, and MongoDB** as the vector database, using **LangChain** and **Langfuse**. Enhanced the project with an AI-based visualization of the codebase to **map functionalities, function flows, and file relationships**, supporting in-depth repository Q&A.
 - Developed an agent model for text2sql with **prompt optimization using DSPy** and LangChain, employing SQL query execution (EX) and exact match (EM) metrics.
 - Configured and **benchmarked the LLM** service (llama-3.1) to generate natural SQL responses, comparing performance with existing models.
 - **Modularized an image processing package into specialized components**, including face matching, ID card photo blur detection, face liveness verification, ID card pen edit detection, tampering detection, and classification.
- Jan 2024 - Jul 2024 **PT Toyota Motor Manufacturing Indonesia** Karawang
Python, AI Developer – Full Time
- Developed an **AI visual inspection system** for the Instrument Panel (IP) Module using **YOLOv8 for object detection**, identifying specifications/parts. **Integrated various devices**: scanner, arduino & limit/switch, and multiple IP cameras.
 - Developed an **AI visual inspection system** for the chassis line, including additional integration with a camera system **connected to a universal robot and OPC Server**.
 - Conducted research and comparative analysis for the **Maintenance Repair Assistant** project.
 - Evaluated **Microsoft Copilot Studio**, providing knowledge through URLs, SharePoint, AI plugins for another task, and external API integration.
 - Implemented **AWS serverless services**: S3 Bucket for data storage, AWS Bedrock for fundamental LLM and embedding models, Amazon OpenSearch Index for vector database functionality, AWS Lambda for function management, API Gateway for API management, EC2 for deploying applications using Docker and Streamlit.
 - **Self-development** using: LLaMA 3 as the base model for LLM, ChromaDB as vector databases, LangChain for orchestration.
- Feb 2024 - Jun 2024 **Skilvul** Remote
AI & Data Science Mentor – Part Time
- Mentor for IBM SkillsBuild for AI & Cybersecurity Bootcamp Program Kampus Merdeka Batch 6 with responsibility:
- Giving live video mentoring via Zoom and chat via Discord for a group of students
 - Collaborate with Technical Lead to make sure learning outcomes are achieved
 - Becoming a facilitator for students in the mentoring or 1 on 1 session
 - Providing guidance, advice, feedback, and support to students in following the program
 - Facilitating students while they are doing their work assignments and final project
 - Assist students from Senior High School, Vocational Students, College Students, Fresh Graduates, and already have a job from diverse backgrounds in developing digital products
- Jun 2023 - May 2024 **Binus Center** Remote
Python and Data Science Mentor – Part Time
- Assist students from Senior High School, Vocational Students, College Students, Fresh Graduates, and already have a job from diverse backgrounds in developing digital products.
 - **Curriculum Development**. Collaborated in designing an effective Python curriculum, tailored to meet both beginner and intermediate students' needs.
 - **Hands-on Guidance**. Led interactive coding workshops and live demonstrations, emphasizing problem-solving techniques, algorithm design, and debugging practices.
 - **Project-Based Learning**. Guided students through project-based tasks, such as building small applications and implementing Python scripts for real-world applications, which enhanced their practical understanding and prepared them for industry roles.

Aug 2021 - Nov 2023	PT Indonesia Indicator <i>Data Scientist, AI Developer – Full Time</i>	South Jakarta
• Performed fine-tuning of the Large Language Model Llama 2 (an alternative to GPT) to align with the company's specific requirements, successfully reducing subscription cost for GPT. • Implemented distributed training and quantization techniques for fine-tuning and inference of the LLM model, effectively reducing resource consumption. • Built a Face Recognition engine using Python and FaceNet, utilized for recognizing faces of political figures, activists, and others. • Developed a system for detecting various types of tuna (albacore, yellowfin, bluefin, etc.) using YOLOv4. Further implemented instance segmentation using Detectron for more precise detection of objects (fish and humans). • Applied Multi-Task Models to develop a Sentiment and Emotion Analysis engine using PyTorch. • Created a classification engine to assess the negativity value of text labels using Python and TensorFlow, enabling in-depth analysis of negative labels. • Developed a near real-time engine for converting speech into text using Transformers wav2vec2. • Created a Virtual Assistant for the Intelligence Media Analytics product, enabling interaction via chat or voice commands. Integrated technologies included wake word detection, speech-to-text conversion, and intent classification for executing and responding to queries. • Applied ONNX conversion to several NLP encoder models, speeding up the inference process by approximately 2x. • Logged all experiments into ML-Flow, ensuring comprehensive documentation of experiment parameters and metrics; facilitated easy tracking of progress and analysis of results. • Integrated all experiments with Apache Airflow, optimizing regular training processes by automating scheduling; increased efficiency by reducing manual intervention and eliminating errors.		

EDUCATION

2017-2021	Telkom Institute of Technology <i>Bachelor of Computer Science; Minor in Data Science</i>	Purwokerto, Central Java
• Cumulative GPA: 3.85/4.00 • Laboratory Assistant of Database Modelling, Programming Algorithm, and Algorithm & Data Structure, with responsibility: ○ Assisting lecturers in teaching both the theoretical and practical aspects. ○ Supporting lecturers in conducting lectures and facilitating practical sessions for students enrolled in the Informatics and Software Engineering Study Program. ○ Collaborating with lecturers to develop and update instructional materials for multiple courses, ensuring effective delivery of course content. • Member, Student Executive Board (BEM KEMA), Ministry of Research and Technology • Award of Hak Kekayaan Intelektual for Kombinasi Jaringan Saraf Tiruan dan Particle Swarm Optimization, Prediksi Jumlah Produksi Air PDAM		

RESEARCHS AND PUBLICATIONS

- [Combination of Backpropagation Neural Network and Particle Swarm Optimization for Prediction of Water Production in Municipal Waterworks](#) (SINTA 2)
- [Pengolahan Citra untuk Membedakan Ikan Segar dan Tidak Segar Menggunakan Convolutional Neural Network](#) (SINTA 4)

CERTIFICATIONS

- [TensorFlow Developer Professional Certificate](#) – DeepLearning.AI in Coursera
 - [Computer Vision Fundamentals with Google Cloud](#) – Google Cloud in Coursera
 - Database Programmer – Badan Nasional Sertifikasi Profesi
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SKILLS

Machine Learning, Deep Learning, Natural Language Processing, Large Language Model, Computer Vision, Object Detection, Python, Pytorch, Streamlit, MongoDB, Elasticsearch, Kafka, Transformers, Airflow, MLFlow, AWS, Ubuntu, DSPy, LangChain, Langfuse, MongoDB, Locust